The Geological Time Scale is best remembered hierarchically: Eon → Era → Period → Epoch → Age

The International Commission on Stratigraphy (ICS) is the authoritative body for the international geological time scale; the current chart was updated in June 2026.

Geological Time Scale — From Earth's Beginning to Present

1. PRECAMBRIAN — about 4567 Ma to 538.8 Ma

Eon	Era	Approximate Time
Hadean	—	4567–4031 Ma
Archean	Eoarchean	4031–3600 Ma
	Paleoarchean	3600–3200 Ma
	Mesoarchean	3200–2800 Ma
	Neoarchean	2800–2500 Ma
Proterozoic	Paleoproterozoic	2500–1600 Ma
	Mesoproterozoic	1600–1000 Ma
	Neoproterozoic	1000–538.8 Ma

Important: The Hadean begins at about 4.567 billion years ago, the age now formally used by ICS.

2. PHANEROZOIC EON — 538.8 Ma to Present

This is the most important part for UPSC because it contains the Paleozoic, Mesozoic and Cenozoic Eras.

A. PALEOZOIC ERA — 538.8 to 251.9 Ma

Period	Approximate Time
Cambrian	538.8–486.9 Ma

Ordovician 486.9–443.1 Ma

Silurian 443.1–419.6 Ma

Devonian 419.6–358.9 Ma

Carboniferous 358.9–298.9 Ma

Permian 298.9–251.9 Ma

Easy sequence:

Cambrian → Ordovician → Silurian → Devonian → Carboniferous → Permian

Memory:

C O S D C P

3. MESOZOIC ERA — 251.9 to 66.0 Ma

Often called the Age of Reptiles.

Period Approximate Time

Triassic 251.9–201.4 Ma

Jurassic 201.4–143.1 Ma

Cretaceous 143.1–66.0 Ma

Easy sequence:

Triassic → Jurassic → Cretaceous

Memory: T–J–K

The Mesozoic ends with the major Cretaceous–Paleogene (K–Pg) mass extinction around 66 Ma.

4. CENOZOIC ERA — 66.0 Ma to Present

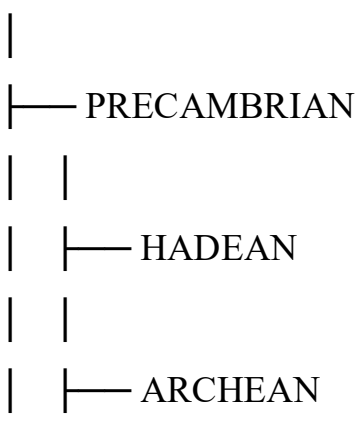
Often called the Age of Mammals.

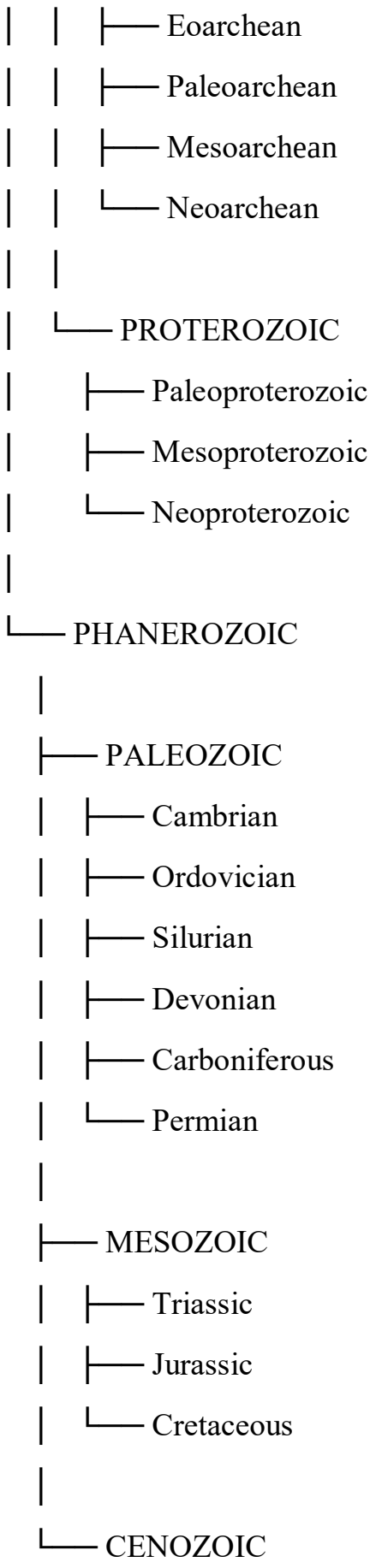
Period	Epochs	Approximate Time
Paleogene	Paleocene	66.0–56.0 Ma
	Eocene	56.0–33.9 Ma
	Oligocene	33.9–23.04 Ma
Neogene	Miocene	23.04–5.333 Ma
	Pliocene	5.333–2.58 Ma
Quaternary	Pleistocene	2.58–0.0117 Ma
	Holocene	0.0117 Ma–Present

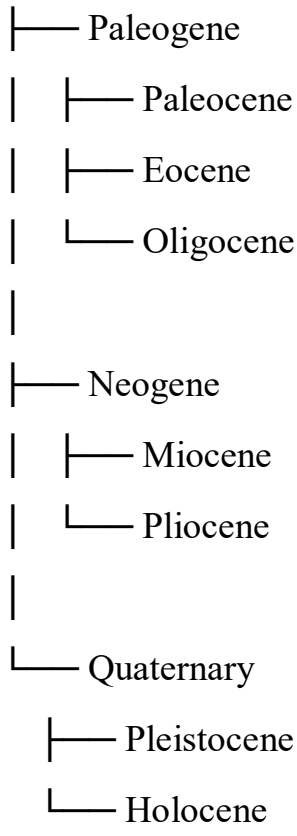
The current ICS chart places the Paleogene, Neogene and Quaternary within the Cenozoic.

 THE COMPLETE UPSC MEMORY TREE

EARTH




Eoarchean
Paleoarchean
Mesoarchean
Neoarchean
PROTEROZOIC
Paleoproterozoic
Mesoproterozoic
Neoproterozoic
PHANEROZOIC
PALEOZOIC
Cambrian
Ordovician
Silurian
Devonian
Carboniferous
Permian
MESOZOIC
Triassic
Jurassic
Cretaceous
CENOZOIC



⭐ One-line chronology to memorise

Hadean → Archean → Proterozoic → Phanerozoic → Paleozoic → Mesozoic → Cenozoic → Quaternary → Holocene

 One important UPSC point

Do not mix up Eon, Era, Period, Epoch and Age:

Eon is the largest → Era → Period → Epoch → Age is smaller.

For example:

Phanerozoic (Eon) → Mesozoic (Era) → Jurassic (Period) → Epoch → Age

The detailed ICS chart goes even further into individual epochs and ages, particularly within the Phanerozoic.

For your students, I would recommend learning the broad hierarchy first and only then adding the Epochs/Ages. That prevents the geological time scale from becoming a confusing list of hundreds of names.

what is Ma as mentioned in the above time zones?

Buddy, Ma means “mega-annum”, i.e. million years ago.

Very simple:

1 Ma = 1 million years ago

10 Ma = 10 million years ago

100 Ma = 100 million years ago

1,000 Ma = 1 billion years ago

4,567 Ma = 4.567 billion years ago

So, when we say:

Cretaceous: 143.1–66.0 Ma

it means the Cretaceous Period began about 143 million years ago and ended about 66 million years ago.

One small terminology point

Strictly speaking:

Ma = million years ago when used on geological time scales.

Ga = billion years ago.

Thus:

4,567 Ma = 4.567 Ga = approximately 4.567 billion years ago.

For students, I would simply teach:

Ma = Million years ago; Ga = Billion years ago.

what is ICS

Buddy, ICS stands for the International Commission on Stratigraphy.

 What is ICS?

The International Commission on Stratigraphy (ICS) is an international scientific body that develops and maintains the Geological Time Scale—the standard framework used to divide Earth's history into:

Eons → Eras → Periods → Epochs → Ages

It is a commission of the International Union of Geological Sciences (IUGS).

Why is ICS important?

ICS establishes internationally accepted boundaries for geological time. For example, it determines where:

Cambrian begins and ends

Jurassic begins and ends

Cretaceous begins and ends

Pleistocene begins and ends

Holocene begins and continues to the present

It also identifies Global Boundary Stratotype Sections and Points (GSSPs)—the internationally agreed physical reference points used to define many geological boundaries.

 Simple student-friendly definition

ICS is the international scientific body responsible for organising and standardising Earth's geological history into a globally accepted Geological Time Scale.

RIVER FLOW FROM MOUNTAINS TO PLAINS AND DELTA

1. The River Begins in the Mountains

Most major rivers originate in mountainous or elevated regions, where rainfall, snowfall, glaciers and springs provide water.

Examples:

Ganga — Himalayan region

Indus — Tibetan/Himalayan region

Brahmaputra — Tibetan Plateau/Himalayas

Godavari — Western Ghats

Narmada — Amarkantak region

At the source, the river generally has: Steep gradient, Narrow channel, High gravitational energy, Rapid flow, Strong erosive power, Large angular rock fragments in its load

Therefore, the upper course is predominantly an erosional zone.

Simple principle

Steep slope + gravitational energy + rapid flow → strong erosion

The river cuts downward into the landscape and develops:

V-shaped valleys, Gorges, Canyons, Rapids, Waterfalls, Potholes

2. What Does the River Erode?

As the river flows downhill, it attacks both its bed and banks.

The material removed from the landscape becomes the river's load.

This load is commonly called detrital material or sediment.

Detrital material includes:

Large boulders, Cobbles, Pebbles, Gravel, Sand, Silt, Clay, Mineral particles, Soil, Organic debris (Dry leaves, dead insects, Dead animals etc)

Thus, Mountain rocks → weathering → erosion → river load → transportation downstream

3. Detrital Material: The River's Moving Load

A very important UPSC concept is that not all sediment travels in the same way.

A. Traction-Large particles such as: boulders, cobbles, large pebbles are rolled or dragged along the river bed.

Large particles → bed → rolling/sliding

B. Saltation-Smaller particles, particularly sand and small pebbles, may bounce or hop along the river bed.

Particle → lifted → falls → lifted again

C. Suspension

Very fine particles such as: silt, clay remain suspended in the flowing water. This is particularly important during floods when rivers carry enormous quantities of suspended sediment.

D. Solution-Some minerals dissolve chemically in the water and are transported in dissolved form.

Therefore, River load = Bed load + Suspended load + Dissolved load

4. What Happens to the Detrital Material?

As the river moves from mountains towards plains, the character of the river changes.

Upper Course	Middle Course	Lower Course
Steep	Gentler gradient	Very gentle gradient

Upper Course	Middle Course	Lower Course
gradient		
High velocity	Moderate/high discharge	Relatively lower energy per unit slope
Vertical erosion	Lateral erosion increases	Deposition dominates
Large angular load	Mixed sediment	Fine sediment increasingly important
Narrow valley	Wider valley	Broad floodplain
V-shaped valley	Meanders	Meanders, floodplains, levees, delta

Important: These are tendencies, not rigid boundaries. Erosion, transportation and deposition can occur simultaneously throughout a river.

5. From Mountain Valley to Plains

When the river descends from mountainous terrain towards a relatively gentle-gradient plain, an important change occurs.

The river: becomes wider, carries more water because of tributaries, develops a larger discharge, loses some of its ability to transport the coarsest material, begins to erode laterally, develops meanders, deposits sediment on its floodplain.

This is the transition from a predominantly erosional landscape towards a landscape of erosion + transportation + deposition.

6. Why Does a River Start Meandering?

This is one of the most important concepts.

When a river flows over a gentle gradient, it does not necessarily travel in a straight line. Small irregularities in the river channel cause differences in velocity.

A bend begins to develop. Once a bend develops: faster flow → outer bank → erosion while, slower flow → inner bank → deposition

This reinforces the bend.

7. Meander: Erosion and Deposition Working Together

A meander is a sinuous or looping bend in a river channel.

It is important to understand that a meander is produced by both erosion and deposition.

Outer bend-Water is relatively faster and has greater erosive energy.

Therefore, Outer bend → erosion → cut bank/river cliff

Inner bend-Water is relatively slower and loses some transporting capacity.

Therefore, Inner bend → deposition → point bar/slip-off slope

This produces the characteristic asymmetry of a meander.

Remember: Outside of bend = Erosion, Inside of bend = Deposition

This is an excellent Prelims memory rule.

8. How Does a Meander Grow?

Consider a river initially flowing almost straight.

Stage 1 — Small irregularity-A slight irregularity develops in the channel.

Stage 2 — Unequal velocity-Flow becomes faster along one side and slower along the other.

Stage 3 — Outer-bank erosion-The faster flow attacks the outer bank.

Stage 4 — Inner-bank deposition-Sediment is deposited on the inner bank.

Stage 5 — Bend becomes larger-Erosion and deposition reinforce the curvature.

Stage 6 — Meander migrates-The meander gradually moves laterally across the floodplain.

Thus, Erosion + deposition → enlarging meander → lateral migration → floodplain development

9. Helicoidal Flow (shaped, coiled, or moving like a helix)

A slightly advanced but important concept is helicoidal flow.

Water does not simply move downstream in a perfectly straight horizontal line. Within a meander, the flow has a spiral or corkscrew-like component.

This helps transfer sediment: Outer bank → erosion

Sediment transported across channel → inner bank

Inner bank → deposition

Thus, helicoidal flow helps explain the continuing development and migration of meanders.

10. Oxbow Lake: When a Meander Is Cut Off

If a meander becomes increasingly pronounced, the two sides of its neck may become very close.

During a flood, the river may cut through the narrow neck and take a shorter route. The old loop is abandoned.

Deposition eventually blocks the old entrances.

The abandoned loop becomes an: Oxbow Lake

Sequence: Meander → Neck becomes narrow → Flood/cut-through → New shorter channel → Deposition closes old channel → Oxbow lake. Eventually the oxbow lake may fill with: silt, clay, organic matter, vegetation and gradually become a wetland or marsh.

11. Deposition of Organic Matter

This is an important addition to the normal discussion of river sediments.

A river does not carry only mineral material. It also transports organic matter, such as: leaves, twigs, roots, dead plants, fragments of vegetation, animal remains, other biological debris

During normal flow, much of this material may continue downstream.

But when water velocity decreases—particularly during: floods, overbank flow, formation of wetlands, oxbow lakes, backwaters, deltaic environments, organic matter can accumulate.

Why?

It is because stagnant or slow-moving water has less capacity to transport material.

Therefore, Slow water → deposition of sediment + accumulation of organic matter

Over long periods, organic-rich deposits can contribute to the development of marshes, wetlands and organic-rich soils.

12. The Formation of Plains

One of the most important consequences of river deposition is the development of alluvial plains.

An alluvial plain is a relatively flat area built largely by sediments deposited by rivers.

The river repeatedly: erodes material upstream, transports it downstream, deposits it when energy decreases, floods its banks,

shifts its channel, spreads sediment across the surrounding land.

Over thousands of years, these processes can produce an extensive plain.

13. Floodplain Formation

A floodplain is the relatively flat land adjacent to a river that is periodically affected by flooding and sediment deposition.

Floodplain formation involves both: Lateral erosion, Meanders erode the valley sides and migrate laterally.

Flood deposition-During floods, water spills over the banks and loses velocity. The sediment carried by the floodwater is deposited across the surrounding land.

Therefore, Lateral erosion + channel migration + repeated flood deposition = floodplain development

14. Why Is the Floodplain Fertile?

Floodwater often carries: silt, clay, minerals, nutrients, organic matter

When floodwater spreads over the plain, it deposits these materials. Repeated deposition can create fertile alluvial soils.

This explains why some of the world's great agricultural civilizations developed along major rivers.

Examples-Ganga–Brahmaputra Plain, Nile Valley, Indus Plain, Mississippi floodplain, Yangtze floodplain

15. Natural Levees

During flooding, water spills over the river bank. The moment water leaves the confined channel, its velocity decreases.

The coarser and heavier sediments tend to settle relatively close to the channel. Repeated flooding gradually builds low ridges along the river banks.

These are called: Natural Levees

Simple sequence- Flood → water leaves channel → velocity decreases → sediment deposited → repeated floods → natural levee

16. Why Does Finer Sediment Travel Farther?

This is a very useful analytical point.

When floodwater leaves the river channel: Velocity decreases. Therefore, sediment settles according to size.

Near the river: Coarser material such as: sand, coarse silt is deposited relatively quickly.

Farther from the river: Finer material such as: fine silt, clay can remain suspended for longer and therefore may be deposited farther away.

Thus, a floodplain can show a depositional gradient from relatively coarse material near the channel to finer material farther away.

17. Point Bars

On the inner side of a meander, deposition creates a gently sloping accumulation of sediment known as a: Point Bar / Meander Bar. Therefore, Outer bend → cut bank, Inner bend → point bar

This is one of the easiest river-landform relationships to remember.

NCERT also identifies point bars/meander bars as depositional features associated with meanders.

18. From Floodplain to Delta- As the river approaches the sea, its gradient becomes extremely gentle. The river enters a much larger water body. Its velocity and transporting capacity may decrease substantially.

Consequently, Sediment begins to accumulate at the river mouth. If sediment accumulation exceeds its removal by waves, tides and currents, a delta may develop.

19. How Is a Delta Formed?

The process can be understood in stages.

Stage 1 — River carries enormous sediment load. Much of the sediment has originated from erosion in the upper and middle courses.

Stage 2 — River enters sea/lake. The river suddenly enters a much larger water body.

Stage 3 — Velocity decreases. The river loses some of its transporting capacity.

Stage 4 — Sediment is deposited. Sand, silt and clay accumulate at or near the river mouth.

Stage 5 — River channel becomes obstructed. Continued deposition creates new sedimentary surfaces.

Stage 6 — River divides. The river finds multiple pathways through the accumulated sediment. These channels are called, Distributaries

Stage 7 — Delta grows-Continued deposition causes the delta to extend towards the sea.

20. Why Does the River Divide into Distributaries?

This is an important conceptual question.

At the river mouth (A Point at which the river joins the sea example, Mouth of Ganga): Deposition → sediment accumulation → obstruction of original channel → water seeks alternative routes

Therefore, the river may split into several smaller channels. These are distributaries.

Remember the distinction:

Tributary-Joins the main river.

Distributary-Leaves/diverges from the main river, particularly in deltaic regions.

21. Delta Plain-The land area occupied by the distributaries and their associated deposits is called the: Delta Plain. It is essentially an extension of the depositional floodplain into the river-mouth environment.

NCERT describes the floodplains within deltas as delta plains.

22. Why Doesn't Every River Form a Delta?

This is an excellent UPSC /PSC, analytical question.

A delta requires a favourable balance between: Sediment supply > sediment removal. A delta is favoured when: river carries abundant sediment, coastal currents are relatively weak, wave action does not rapidly remove deposited sediment, tidal removal is not strong enough to disperse the sediment, the coastline and nearshore conditions allow sediment accumulation.

If waves, tides and coastal currents remove sediment as rapidly as the river supplies it, a prominent delta may not develop.

23. Delta vs Estuary

This distinction is frequently tested.

Delta	Estuary
Deposition dominates	Sediment is often carried away/reworked
River divides into distributaries	Usually a funnel-shaped tidal river mouth
Sediment accumulates at mouth	Strong tidal influence

Builds land seaward delta	Does not necessarily build a depositional
Ganga-Brahmaputra	Narmada
Godavari	Tapti
Krishna	Many strongly tidal/coastal settings

Indian examples

Delta-forming rivers: Ganga, Brahmaputra, Mahanadi, Godavari, Krishna, Cauvery

Major estuarine examples: Narmada, Tapti

24. Complete River Journey — The Big Picture

Think of the entire system as: MOUNTAINS- Weathering + erosion Detrital Material-Boulders → pebbles → gravel → sand → silt → clay

TRANSPORTATION-Traction → Saltation → Suspension → Solution

MIDDLE COURSE-Lateral erosion increases

MEANDERS - Outer bank erosion, Inner bank → deposition

FLOODPLAIN-Lateral migration + flood deposition

ALLUVIAL PLAIN-Repeated sediment accumulation

LOWER COURSE-Gentle gradient + declining transporting capacity

RIVER MOUTH-Sediment deposition

DISTRIBUTARIES-River divides

DELTA-New land is progressively built outward.

This is why river plains and deltas can become ecologically productive environments, supporting wetlands, agriculture, fisheries and diverse ecosystems.